KODLARIM

TESTBENCH.SV kodları

```
`timescale 1ns / 1ps
module tb_mux_4to1_8bit;
   reg [7:0] d0;
   reg [7:0] d1;
   reg [7:0] d2;
   reg [7:0] d3;
   reg [1:0] sel;
   wire [7:0] y;
   mux_4to1_8bit uut (
      .d0(d0),
      .d1(d1),
      .d2(d2),
      .d3(d3),
      .sel(sel),
      .y(y)
   );
   initial begin
      $dumpfile("dump.vcd");
      $dumpvars(1, tb_mux_4to1_8bit);
      d0 = 8'hAA;
      d1 = 8'h55;
      d2 = 8'hCC;
      d3 = 8'h33;
      sel = 2'b00; #10;
      sel = 2'b01; #10;
      sel = 2'b10; #10;
      sel = 2'b11; #10;
      d0 = 8'h11;
      d3 = 8'hEE;
```

```
        sel = 2'b00; #10;
        sel = 2'b11; #10;
        $finish;
    end
    initial begin
        $monitor("Zaman=%0t | sel=%b | d0=%h, d1=%h, d2=%h, d3=%h | Çıkış (y)=%h",
                 $time, sel, d0, d1, d2, d3, y);
    end
endmodule
```

DSİGN.SV kodları

```
module mux_4to1_8bit (
    input  wire [7:0] d0,
    input  wire [7:0] d1,
    input  wire [7:0] d2,
    input  wire [7:0] d3,
    input  wire [1:0] sel,
    output reg  [7:0] y
);
always @(*) begin
    case (sel)
        2'b00: y = d0;
        2'b01: y = d1;
        2'b10: y = d2;
        2'b11: y = d3;
        default: y = 8'h00;
    endcase
end
endmodule
```

KOD ÇIKTISI:

```
# KERNEL: Zaman=0 | sel=00 | d0=aa, d1=55, d2=cc, d3=33 | Ä‡Ä±kÄ±ÅŸ (y)=aa
# KERNEL: Zaman=10000 | sel=01 | d0=aa, d1=55, d2=cc, d3=33 | Ä‡Ä±kÄ±ÅŸ (y)=55
# KERNEL: Zaman=20000 | sel=10 | d0=aa, d1=55, d2=cc, d3=33 | Ä‡Ä±kÄ±ÅŸ (y)=cc
# KERNEL: Zaman=30000 | sel=11 | d0=aa, d1=55, d2=cc, d3=33 | Ä‡Ä±kÄ±ÅŸ (y)=33
# KERNEL: Zaman=40000 | sel=00 | d0=11, d1=55, d2=cc, d3=ee | Ä‡Ä±kÄ±ÅŸ (y)=11
# KERNEL: Zaman=50000 | sel=11 | d0=11, d1=55, d2=cc, d3=ee | Ä‡Ä±kÄ±ÅŸ (y)=ee
# RUNTIME: Info: RUNTIME_0068 testbench.sv (40): $finish called.
# KERNEL: Time: 60 ns,  Iteration: 0,  Instance: /tb_mux_4to1_8bit,  Process: @INITIAL#21_0@.
# KERNEL: stopped at time: 60 ns
```